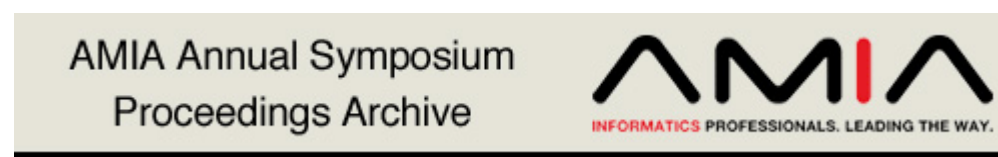


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Balancing Performance and Interpretability: Selecting Features with Bootstrapped Ridge Regression

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Abstract

Informaticists sometimes attempt to predict chronic healthcare events that are not fully understood. The resulting models often incorporate copious numbers of predictors derived across diverse datasets. This approach may yield desirable performance characteristics, but it sacrifices interpretability and portability. The Bootstrapped Ridge Selector (BoRidge) offers a tool to balance performance with interpretability. Compared to two modern feature selection methods, Bootstrapped LASSO regression (BoLASSO) and a minimal-redundancy-maximal-relevance selector (mRMR), the BoRidge bested them for binary classification on artificially generated data (sensitivity: 0.83, specificity: 0.72) versus BoLASSO (sensitivity: 0.1, specificity: 1) and mRMR (sensitivity: 0.69, specificity: 0.69). On a dataset used to validate a published suicide risk prediction model, the BoRidge selected an equally precise model to the publication, with far fewer predictors (114 versus the 1,538 used in the published model). The BoRidge has the potential to simplify classification models for complex problems, making them easier to translate and act upon.

Background

Advances in machine learning have allowed predictive models to permeate more complex clinical and behavioral domains¹⁻³. Enigmatic healthcare outcomes may have many interdependencies and may require predictive models of greater intricacy in both their methods and the number and complexity of predictors used⁴. For example, the risk factors for suicidal thoughts and behaviors have been the subject of frequent study for over fifty years^{5,6}. Clinicians found that a complex interaction between social factors, severity of chronic illness/injury, and behavioral comorbidities preempt suicidal thoughts and behaviors⁷. Clinicians and researchers only recently have retrospectively modeled these events with better accuracy than risk assessment scores. Risk scores generally performed near chance levels in meta-analytic analysis⁸. At the same time, providers seek predictive models that are as readily interpretable as traditional risk models such as the Framingham risk score used in cardiovascular risk prognostication⁹. Models with hundreds or thousands of predictors or those that use deep architectures preclude this interpretation. They may miss opportunities to provide further data-driven insights for clinicians on how to act on high-risk patients. This manuscript will focus on methods for scaling back model complexity caused by a large number of predictors (features), while maintaining performance. This is not to say that large numbers of predictors don't have benefits, such as performance in difficult domains and resistance to calibration drift, but they do have costs^{10,11}.

Large numbers of predictors present a barrier to model portability and implementation. A vast array of predictors can present a formidable burden developing the requisite data feeds. This coupled with the translation of each variable definition between clinical systems and organizations may lead to nuance that requires clinical expertise to disentangle. This kind of complexity may partially explain why the most commonly implemented readmission risk models also tend to be the simplest, i.e., LACE or CORE^{12,13}.

Predictive models are meant to provide clinicians with information that enables them to implement an intervention to prevent a harmful or costly future event. This form of decision support should be interpretable to empower action^{11,14}. Models with a myriad of features present challenges for interpretability¹⁵. Interaction terms and non-linear variable transformations can further complicate interpretability. Model interpretability may not only inform action but may also build the trust of the model's consumers¹⁶. The benefits of smaller and interpretable models have weight that should not be discarded wholesale for the pursuit of performance.

Additionally, models with a large number of features may be vulnerable to faults in upstream systems through dependencies on the numbers and types of variables used for prediction. This exposure to many different kinds of faults may make the maintenance of such models nontrivial. Even simple rule based decision support systems require advanced analytical infrastructure to effectively maintain over time^{17,18}. The maintenance and implementation costs for models with a large number of predictors may add up and hinder information technology personnel and resources. Balancing model performance with other considerations may lead to more resilient and portable predictive models.

One means to limit model complexity from multiple variables is to reduce the model by selectively eliminating predictors. The field of feature selection is well developed and has mature roots in the statistics literature on model selection. A classic method for reducing features is comparison of nested regression models using the likelihood ratio test^{19,20}. This method tests for significant differences in the fit between a parent (full) model and a reduced model. One may choose the reduced model if no significant difference exists. This approach is problematic because it restricts comparison to nested regression models, and relies on the absence of evidence as the basis for selecting the reduced model. Another classic feature selection approach is the use of stepwise selection, which either adds one variable at a time or subtracts one variable at a time. Stepwise selection optimizes based on a specific criteria such as the F-Test statistic or an information criterion such as Akaike's

Information Criterion²¹. However, stepwise selection can lead to overestimation of the selected model coefficients (bias) and has consistency problems²². Researchers have also examined the use of correlation based methods to select the features most correlated with the outcome^{23,24}. This approach can have limited efficacy as correlation methods are sensitive to outliers and skew of the data^{25,26}. Penalized regression methods seek to bias model coefficients toward zero in an effort to reduce the variance of model residuals. There are different methods for optimizing this bias variance tradeoff; the two most common minimize either the L1 norm (LASSO) or the L2 norm (Ridge) of the least squares equation. Penalized regression allows for some variables to experience more shrinkage than others.

Researchers have recently developed more promising methods of feature selection. Researchers took the LASSO a step farther by using resampling techniques to fit multiple LASSO regressions. They then selected the features that are consistently weighted above zero²⁷. This approach is statistically consistent, has faster convergence to asymptotic properties than other feature selection methods, and is robust to the LASSO consistency assumption²⁸. Another modern method is the minimal-redundancy-maximal-relevance criterion (mRMR). mRMR seeks to maximize information and correlation, while penalizing features for redundancy²⁹. In the mRMR scheme, features that do not add enough new information or are not correlated enough with the outcome are eliminated. While there are many existing methods for feature selection, no single method dominates the others, which leads to tradeoffs. Ridge regression has some unique advantages such as its propensity to keep predictors, its intuitive geometric optimization, and its ability to resist calibration drift over time^{10,30}. By combining Ridge regression with resampling techniques such as bootstrapping, one may be able to produce more compact yet resilient models. This work will seek to compare a novel bootstrapped ridge feature selector with other modern techniques to provide evidence for value. The goal of this work is to produce an effective tool for refining predictive models to be compact, interpretable, and portable.

Methods

Generation of Artificial Data To evaluate the different feature selection methods the authors sought to demonstrate effectiveness across two different types of outcomes: continuous regression and binary classification. These outcome types provide coverage of many of the types of events informaticians are trying to predict. The authors also sought to represent different types of predictors to ensure the bootstrapped ridge is able to span a gambit of data generation processes.

To this end, the authors generated two datasets (using R version 3.4.3), one with a continuous outcome and the other with a binary outcome. Each dataset had 50 true predictors and 50 variables uncorrelated variables. The continuous outcome was the sum of the 50 true predictors: five independent Poisson variables, five independent normal variables, five independent gamma variables, ten independent binomial variables, 25 correlated multivariate normal variables, and one independent normally distributed error term. The error term was not included in the data set. Next, the authors extended the data set with 50 variables independent of the outcome that included five independent Poisson variables, five independent normal variables, five independent gamma variables, ten independent binomial variables, and 25 correlated multivariate normal variables. The parameters for each distribution were randomly selected. The authors generated 10,000 observations for each dataset. From the dataset of 10,000 observations, the authors bootstrapped 1,000 samples with 200 observations, 1,000 samples with 400 observations, 1,000 samples with 800 observations and 1,000 samples with 1,600 observations. This sampling process created four groups of data with: two observations to one predictor, four observations to one predictor, eight observations to one predictor, and sixteen observations to one predictor respectively.

The binary outcome dataset was created with similar methods. The main exception being that the outcome was marked as a positive case if that sum was greater than a random percentile of sums for 10,000 observations. The primary data set of 10,000 observations with outcomes was again bootstrapped into 1,000 samples with 200 observations, 1,000 samples with 400 observations, 1,000 samples with 800 observations and 1,000 samples with 1,600 observations.

Bootstrapped Ridge (BoRidge) Implementation The authors sought to build on the well-developed open source platforms offered by Python including the Pandas and Scikit-Learn libraries for their implementation of the BoRidge^{31,32}. Ridge regression does not naturally select features, even though it uses the statistical principle of shrinkage to bias variable coefficients toward zero. To achieve variable selection the authors used two levels of bootstrap resampling to build confidence intervals for each predictor. At the first level, a sample with replacement is drawn from the original dataset, and a Ridge regression is fit. When fitting the regression, the Ridge penalty parameter is tuned to optimize the log loss of the observations not in the bootstrap (out of bag) through a scan of possible values. This process is repeated a number of times (100 is the default) to provide a distribution of point estimates for the variable coefficients. At each first level bootstrap, the same number of bootstrap samples (default is 100) is drawn from the first level sampled data, creating a second level. [Figure 1](#) illustrates the two-level resampling process.

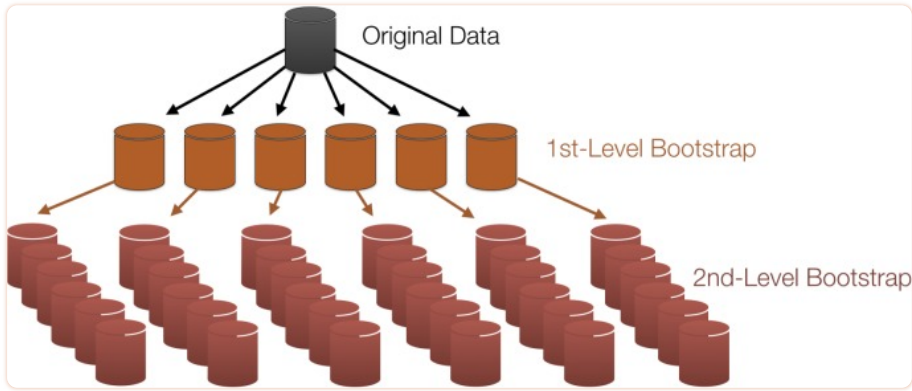
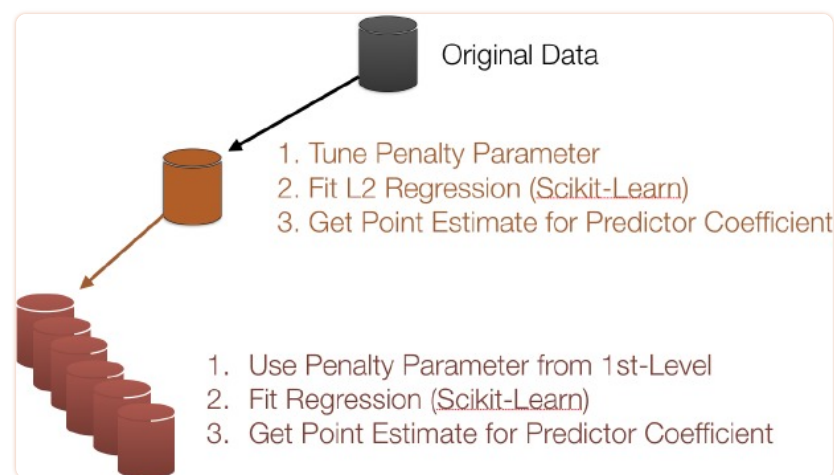


Figure 1.

Example of 2-Level (Double) Bootstrap

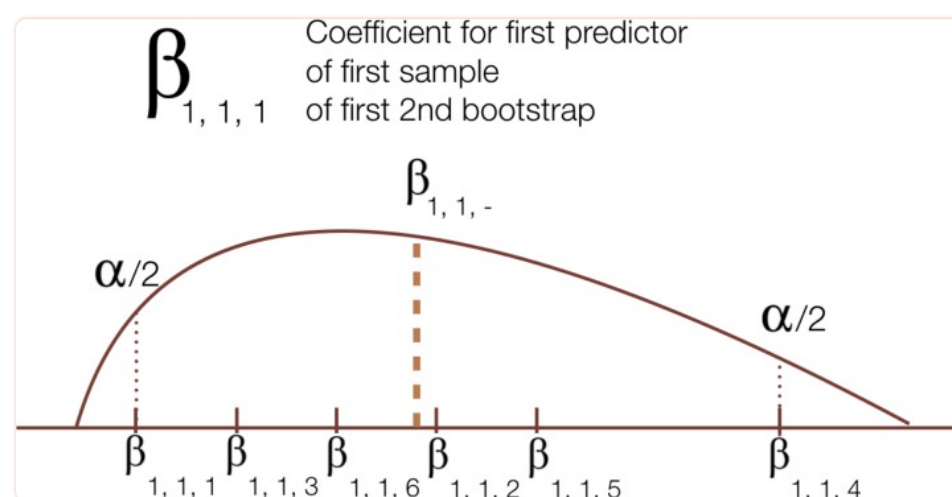
At the second level, a Ridge regression is fit on the second level resampled data where the penalty parameter is inherited from the optimal value found for the first level. [Figure 2](#) provides an illustration of the events that occur at each bootstrap level.



[Figure 2.](#)

Steps Taken For Each Sample

The point estimates for the variable coefficients are stored to form a distribution centered at the point estimate from the first level. In essence, the second level provides an estimate of the variance of the first level. The authors specifically utilized the bootstrap pivot interval for building the confidence intervals of the beta coefficients at the first level³³. This method is nonparametric and uses the empirical distribution of the second level coefficients to build a confidence interval. [Figure 3](#) illustrates how the second level is used for the confidence interval of the first level variable coefficients.



[Figure 3.](#)

Principle Underlying the Bootstrap Pivot Interval

The specific formula used can be referenced in equation 1.

$$\text{Lower Bound} = 2 \times \beta_{1st-Level} - \beta_{2nd-Level}^{(1-\alpha/2)\text{percentile}}$$

$$\text{Upper Bound} = 2 \times \beta_{1st-Level} - \beta_{2nd-Level}^{(\alpha/2)\text{percentile}}$$

Equation 1: Bootstrap Pivot Interval

After calculating the confidence interval for each variable coefficient, the BoRidge tests to see if the confidence intervals contain the null hypothesis. If the interval excludes the null hypothesis, then the BoRidge marks that feature as selected for that first level sample. Variables that are consistently selected across multiple bootstrap samples should be considered more valuable predictors. The number of bootstraps employed for the threshold of variable significance is tunable. Variables found to be significant at or above the threshold number of first level bootstraps should be kept in the model, while the other variables should be discarded.

Evaluation for Artificial Data The authors compared the sensitivity and specificity of the BoRidge at selecting the true predictors in both the linear and binary data sets against an mRMR feature selector³⁴ as well as a BoLASSO feature selector³⁵. The two comparative methods represent two relatively modern feature selection methods. The authors used a soft BoLASSO threshold of 90% of the bootstraps, as suggested by

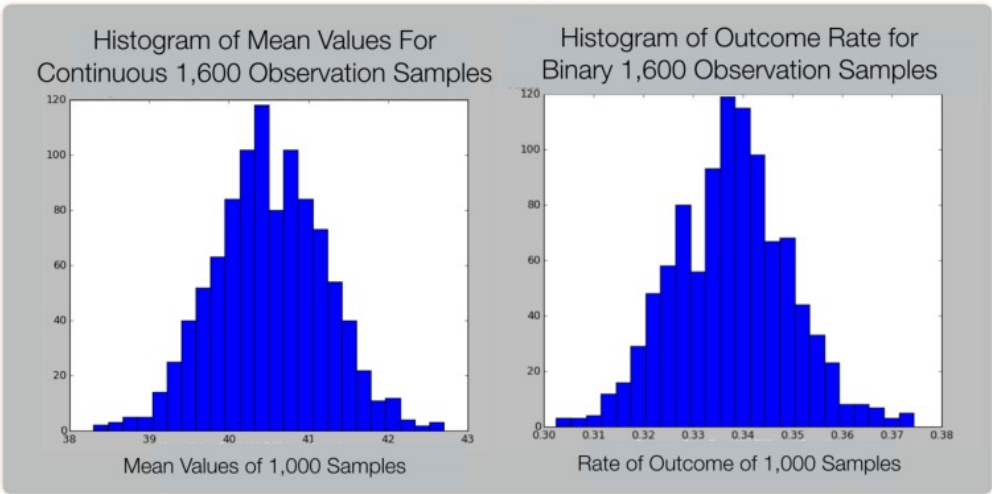
Bach²⁸. For the mRMR package the authors had to pre-specify the number of results to select, leading them to choose the number of true predictors (50). For each set of 1,000 samples the sensitivity and specificity of the variables selected were averaged for all the feature selection methods.

Case Study Data: Suicide Attempt Prediction The authors sought to demonstrate the utility of the BoRidge by having it attempt to reduce a published prediction model. The authors felt that suicide risk prediction made for a good case study because the causes and risk factor for suicide are complex and still the subject of much research. A high-performing model in this space uses over 1,500 features to generate a prediction and is in the process of real world implementation⁸. A more readily interpretable model may encourage uptake by potential end-users while also improving portability to new settings. The authors experimented with the same de-identified data from the Vanderbilt Synthetic Derivative (SD) used to validate Walsh et. al's model⁸. These data were drawn from the SD from 1998-2015 and comprise 3,250 cases of suicide attempt validated through two expert chart review (Kappa ~0.95) and approximately 12,000 controls seen in routine medical care at Vanderbilt University Medical Center. Predictors comprised prior diagnostic codes grouped to Center for Medicare and Medicaid Services Hierarchical Condition Categories, demographics including age, gender, race, documented medications grouped to Anatomic Therapeutic Classification, Level V, and prior healthcare utilization including inpatient, outpatient, and emergency department visits.

Case Study Evaluation The authors contrasted both BoLASSO and mRMR selected models' optimism-adjusted area under the receiver operator characteristic curve (AUC) against that of the BoRidge selected model. The authors also compared the AUC of the reduced random forest models to the full random forest model of Wlash et al. The authors thematically grouped the selected predictors of the BoLASSO, mRMR, and BoRidge models for comparative purposes. Again, mRMR required a pre-defined number of significant covariates to consider. The authors iteratively increased this parameter until the resulting model equaled or bettered the optimism-adjusted AUC of the base-line model from Walsh et. al⁸.

Results

After drawing bootstrap samples from the artificial continuous and the artificial binary outcome data, there were 8,000 total data sets. One set of 1,000 for each observation count (200, 400, 800, and 1,600) for each type of outcome (continuous and binary). The samples varied randomly, forming a distribution of different parameter values. [Figure 4](#) is an example of the distribution of the central moment for both the continuous and binary outcomes for the 1,600 observations count set. Different observation counts sets centered on different values. For example, the binary 400 observations set had an average outcome rate of 0.71. The distribution of each set was generally symmetric and unimodal.



[Figure 4.](#)
Sample Histogram of Artificial Data Distribution

The authors sought to compare the average sensitivity and specificity for both the continuous (linear) outcomes as well as the binary outcomes over the different draws of data. Due to the low variability of the sensitivities and the specificities and the high number of samples, all differences were found to be statistically significant where the dots do not overlap. [Figure 5](#) visualizes the performance characteristics of the three different feature selection methods for continuous data.

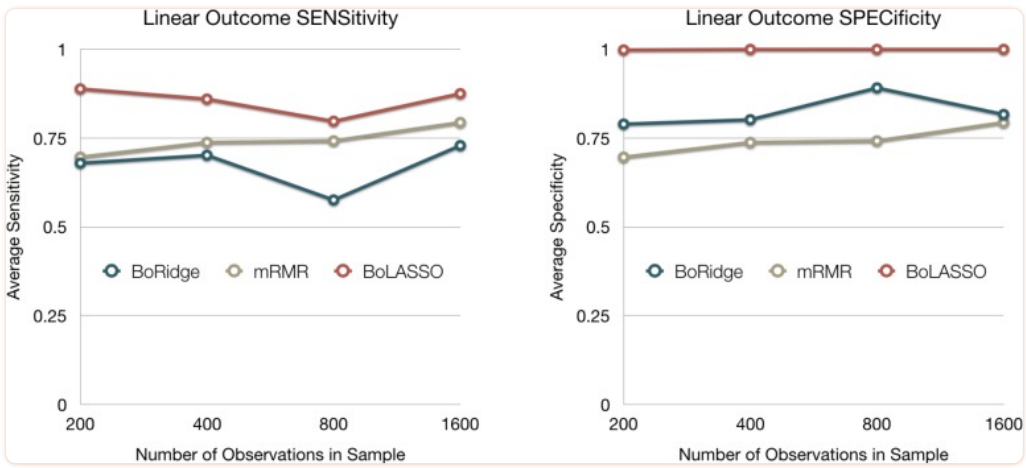


Figure 5.

Artificial Data Evaluation for Average Linear Outcome Sensitivity and Specificity

For the continuous outcome, the BoLASSO feature selector proved superior in both sensitivity and specificity across all sampling levels. The BoLASSO was nearly perfectly specific, yet still was able to find the true predictors. The mRMR algorithm was generally more sensitive than the BoRidge. The BoRidge was able to maintain better specificity than mRMR across different sample sizes. The BoRidge struggled in identifying the binomial and Poisson predictors, which dragged down its sensitivity.

The BoRidge fared better finding the predictors for the binary outcome as shown in Figure 6. The BoRidge bested the average sensitivity across sample sizes, though mRMR was almost as proficient. The BoLASSO struggled to select predictors leading to low sensitivity, but high specificity. The relative specificity of the mRMR selector to the BoRidge was similar for the smaller samples, but the BoRidge diverged at the 1,600 observations set.

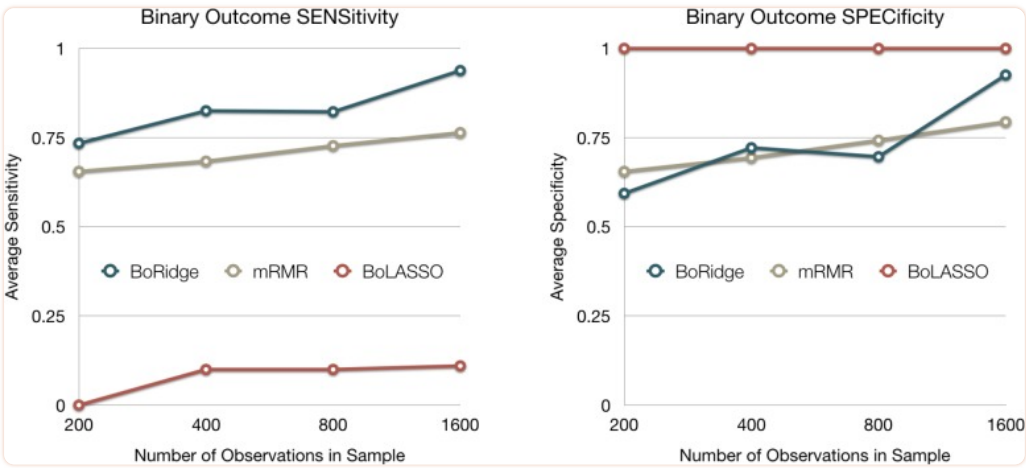


Figure 6.

Artificial Data Evaluation for Average Binary Outcome Sensitivity

The case study data set is made from 15,908 adult patients with a diagnosis code for depression between the years 1998 and 2015. From the total cohort, 3,249 have a documented suicide attempt validated through chart review by a team of psychologists and suicide researchers. The authors extracted 1,538 predictors that accumulated up to 7 days before a suicide attempt. The cohort characteristics are documented in Table 1 with p-values generated from the pooled sample proportions.

Table 1.

Cohort Characteristics for Case Study Evaluation

Characteristic	Cases n=3,250	Control n=12,659	P-Value
Gender			
Male	1,696 (0.52)	5,501 (0.43)	<0.05*
Female	1,534 (0.47)	7,115 (0.56)	<0.05*
Unknown	20 (0.006)	43 (0.003)	0.57
Race			
White	2,706 (0.832)	8,026 (0.634)	<0.05*
Black	375 (0.115)	1,040 (0.081)	<0.05*
Other	69 (0.021)	3593 (0.284)	<0.05*
Age			
Median (SD)	37.1 (13.1)	49.1 (18.1)	

The BoLASSO selected 37 out of 1,538 features that produced a Random Forest model that had an optimism-adjusted area under the receiver operator characteristic curve (AUC) of 0.84. The mRMR selector picked a 197-feature Random Forest model with an optimism-adjusted AUC of 0.86. The BoRidge selected 114 features that produced a Random Forest model with an optimism-adjusted AUC of 0.85.

Due to the number of variables selected by both models, the clinician co-author summarized the predictors into thematic categories. A summary of these themes is presented in [Table 2](#). These themes are labeled as being protective or being risk factors based on a regression fitted with the same features as the Random Forest model.

Discussion

In the artificial datasets, the BoRidge best selected features for the binary outcome. It outperformed both the BoLASSO and mRMR. The BoRidge did not consistently identify the binomial and Poisson predictors for both the continuous outcome and the binary outcome. These predictors generally contributed only a small piece of the total summed outcome. Their relatively small contributions made them difficult to detect, especially when potentially offset by an unobserved random error. The mRMR selector was the most consistent performer of the group averaging around a 0.75 for both sensitivity and specificity regardless of the outcome or sample size. The BoLASSO did very well selecting features for the continuous outcome, but had great trouble identifying predictors for the binary outcome. All of the feature selectors were generally more selective with more data per sample. The BoRidge appears to be the best option for binary outcomes, while the BoLASSO appears to be superior for continuous outcomes.

All methods — BoLASSO (37 predictors), mRMR (197 predictors), and the BoRidge (114 predictors) — greatly reduced the feature space of the predictive model in the case study. The performance of the BoLASSO (AUC of 0.84), mRMR (AUC of 0.86), and BoRIDGE (AUC of 0.85) selected models aligned with the optimism-adjusted AUC reported by Walsh et. al. of 0.84. Reducing the feature space of 1,538 by an order of magnitude or more is a boon for clinicians looking for a brief summation of the most predictive patient risk factors. This reduction is also beneficial for Health IT personnel looking to externally validate and deploy a new model at their institution. Each model variable may require a pipeline to clean, impute, and/or import before it is ready for clinical use. Feature selection can also improve models by eliminating noisy or misleading predictors in the training data. The BoRidge was less parsimonious than the BoLASSO, demonstrating the properties that contribute to Ridge regression's resilience to calibration drift¹⁰. By being less selective, the BoRidge may be a better alternative to the BoLASSO as a hedge against over-fitting. The only thematic category of predictors that was included by the BoLASSO, but excluded by the BoRidge was emergency department utilization. The mRMR selector chose even more covariates than the BoRidge to achieve similar performance. The mRMR selector picked more demographic, behavioral, and utilization covariates than either the BoRidge or the BoLASSO. The BoRidge included more physical comorbidities as competing risks to suicide than mRMR.

As far as other themes, prior poisoning, suggestive of suicidality, was a major risk and is specifically identified as a risk factor in prior research⁵. Severe depression with psychosis was another important risk factor. All methods found the themes of inpatient utilization, history of mental illness, and prior poisoning attempts as important risk factors for suicide. These variables have the potential to inform which patients may require additional care and follow-up. Severe medical illness across categories such as hematologic disease or malignancy, diabetes, and cardiac disease were found to be protective factors, because they act as competing risks to suicide. Other protective factors included increasing age, which is consistent with the literature, as well as medications typical of outpatient care: vaccinations for chronic diseases, anti-hypertensives, anti-hyperlipidemics, and non-narcotic analgesics.

To the authors' knowledge the bootstrapped Ridge regression is a novel feature selection method. The evaluation above suggests that the BoRidge may provide some unique characteristics not represented by other feature selection methods. Even in one of the original implementations of the BoLASSO, its authors had to tune how many bootstraps a feature was considered significant in before the feature was selected²⁸. In the case study, the BoRidge required no tuning (variables were significant in all bootstraps) to achieve the same discrimination as the "soft" BoLASSO model (feature was included 90% of the bootstraps or more). The primary parameter of the BoRidge is the number of bootstraps, where more bootstraps leads to greater consistency of the variables selected for a particular dataset. The authors believe the BoRidge's bootstrap parameter is less arbitrary than the mRMR parameter for the number of significant variables to select. The authors found that model performance could vary greatly depending and how many covariates mRMR was set to find. This means that it is important to have some kind of base line when iteratively using mRMR to select features.

Feature selection has many promising applications especially in a learning health system. With enough data, one can use feature selection to build models from the first principles. This may eliminate issues with the translation of predictive models from one site to another. A site could simply use feature selection to discover the model that fits their local systems, skipping what can be the difficult semantic translation of predictors from one health system to another. Feature selection can also help modelers make their predictions more explanatory by honing in on specific covariates. Some organizations are pairing simpler explanatory models with deep-learning models. This ensemble attempts to pair the most accurate predictions, while also providing clinical insight into the prediction.

Many areas of clinical care, such as suicide prevention, have complex root causes and risk factors. Feature selection is a means to focus the "kitchen-sink" approach to achieve accurate predictions. Feature selection can also make such models more compact and therefore more portable from one site to the next. A key element of portability is the external validation of predictive models in a new environment. Models with large feature sets may be difficult to validate in different clinical systems or settings, because predictors may not be available. Reducing feature sets to the minimum needed for generalization lowers a barrier for external validation.

This study has several limitations. The authors only explored a fairly simplistic linear relationship for the continuous outcome. Some form of polynomial type relationship may better represent realistic data. The case study contained data from one site, which could limit the generalizability of the BoRidge selected model. The cohort for the case study had demographic imbalances, which may have led to non-modifiable risk/protective factors in the predictive model.

The BoRidge represents a unique use of Ridge regression for feature selection. For binary outcomes the BoRidge shows promise as a superior feature selector compared to the BoLASSO and mRMR. The BoRidge selector can greatly reduce feature spaces of some predictive models leading to greater translatability. Feature selection can help optimize models to be more interpretable while still being precise.

Table 2.

Themes of Features Selected by BoLASSO, mRMR, and BoRidge

Theme	BoLASSO	mRMR	BoRIDGE	Factor Type
Demographics				
Race	X	X	X	Risk for Caucasians
Gender		X	X	Risk for Men
Age	X	X	X	Protective
Area Deprivation Index		X		Risk for Low-Income
Behavioral Comorbidities				
Substance Abuse		X	X	Risk
History of Mental Illness	X	X	X	Risk
Prior Poisoning	X	X	X	Risk
Absence of Prior Self-Inflicted Trauma		X		Protective
Absence of Behavioral Medication		X	X	Protective
Presence of Behavioral Medication		X		Risk
Physical Comorbidities				
Cardiac Conditions		X	X	Protective
Endocrinological Conditions			X	Protective
Hematologic Conditions			X	Protective
Vascular Conditions	X	X	X	Protective
Healthcare Utilization				
Inpatient Utilization	X	X	X	Risk
Emergency Room Utilization	X	X		Risk
Outpatient Utilization		X		Risk

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